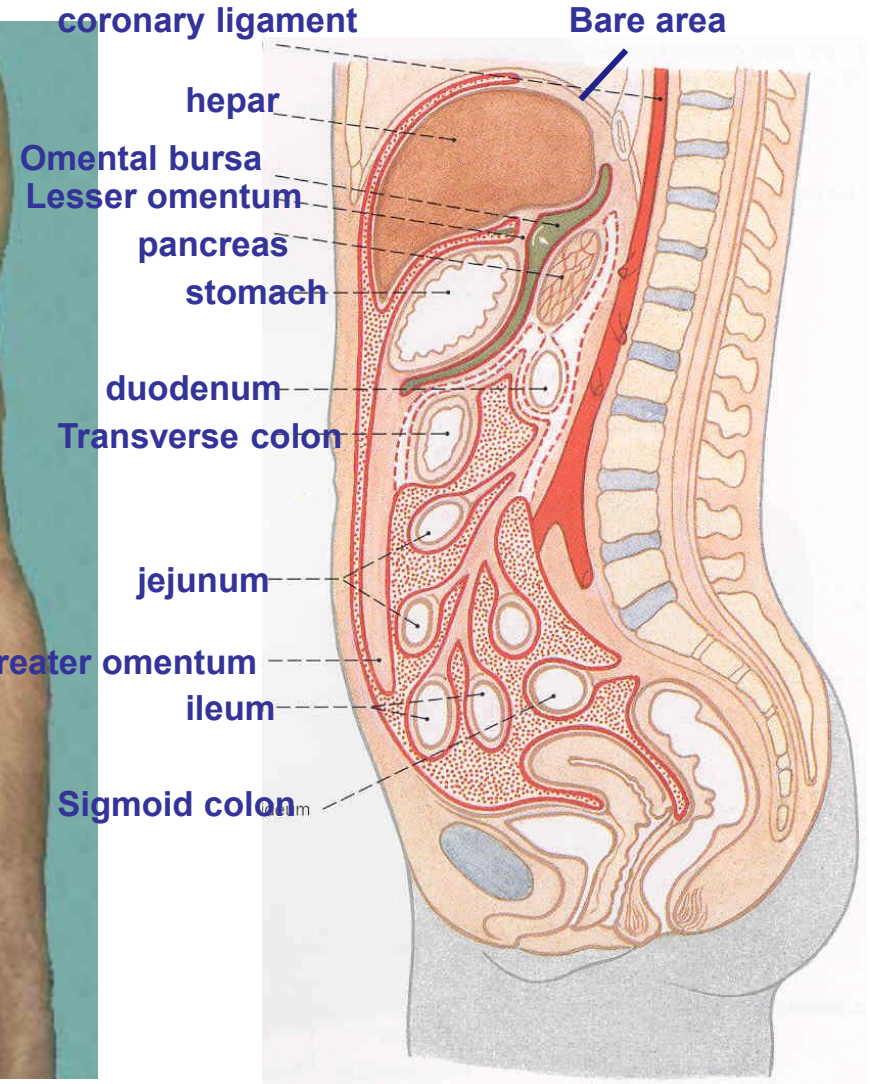
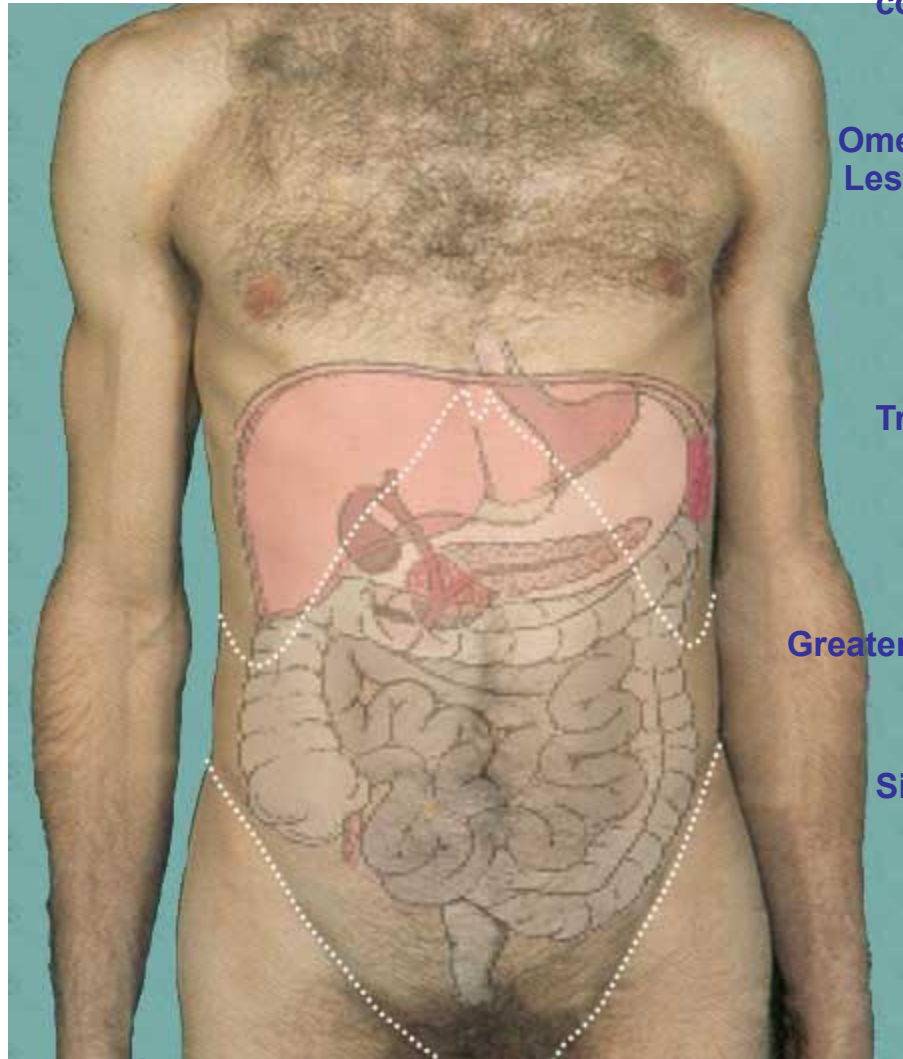


POSITION OF THE LARGE INTESTINE AND RECTUM



ANATOMICAL DIVISIONS

CECUM + vermiform appendix

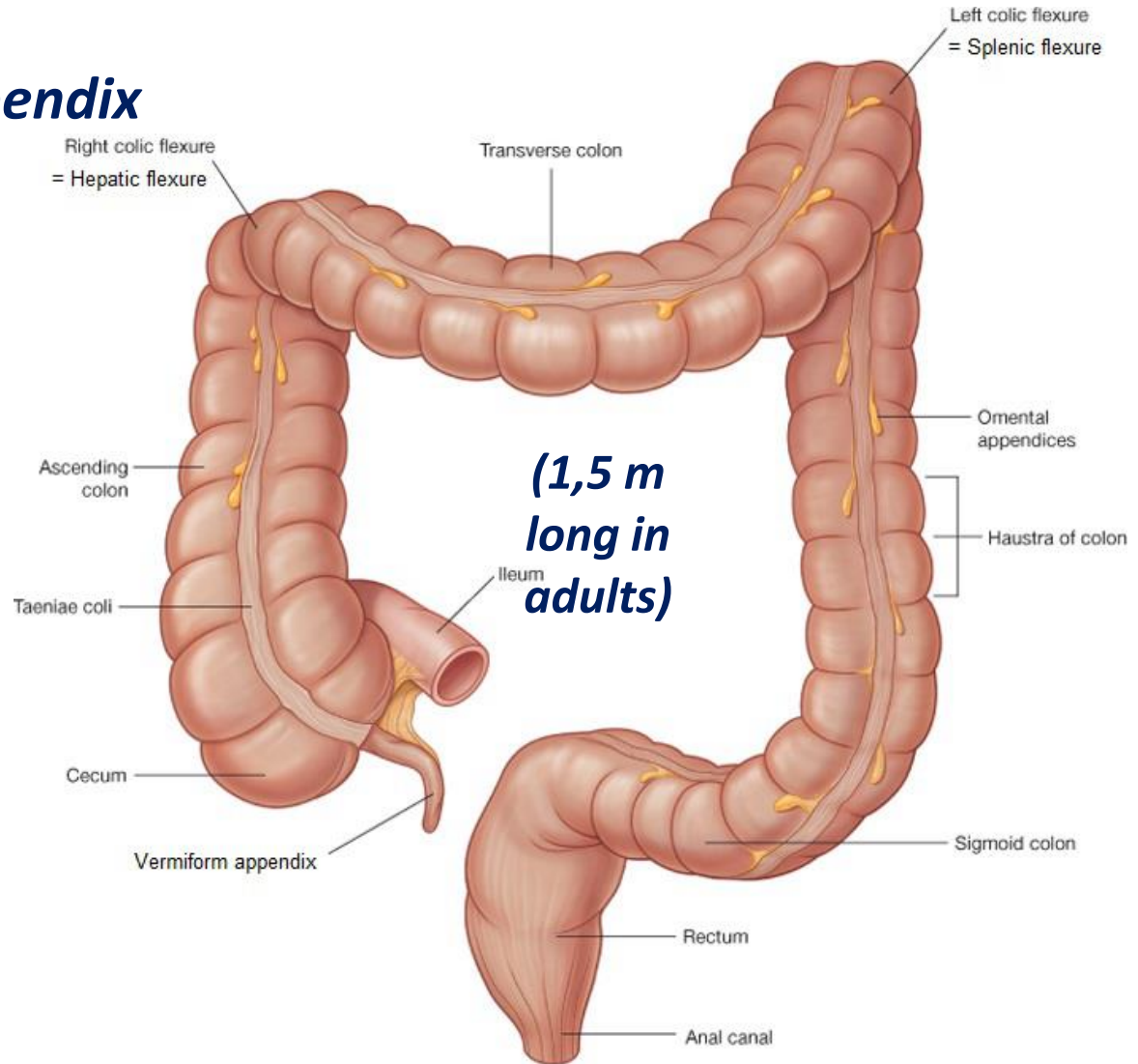
ASCENDING COLON

TRANSVERSE COLON

DESCENDING COLON

SIGMOID COLON

RECTUM



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HOW CAN WE DISTINGUISH THE LARGE AND SMALL INTESTINES?

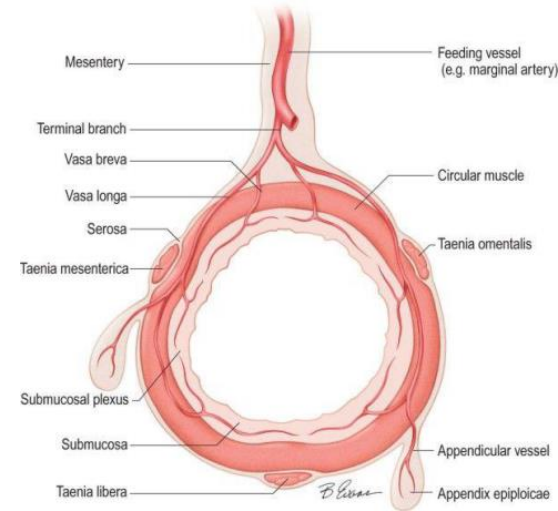
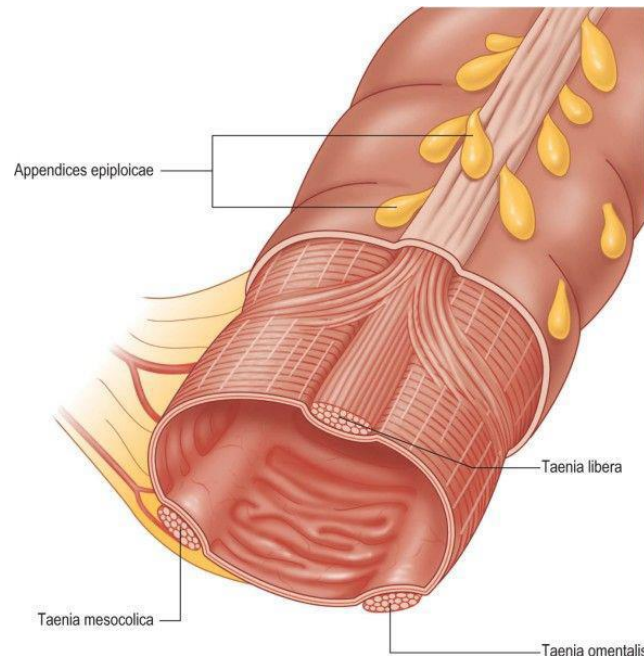
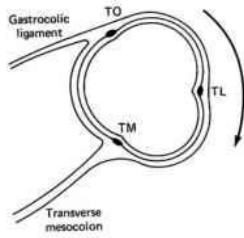
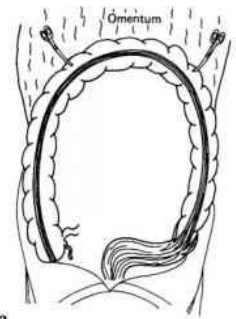
External features

Epiploic appendices - protrusions of the peritoneum filled with areolar /fatty connective tissue

Haustra (plural) - saccular dilations of the wall

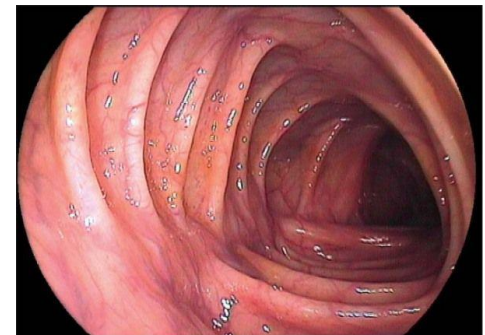
Taeniae (plural) - strong bands of the tunica muscularis (absent from the vermiform appendix, sigmoid colon and rectum)

Omental
Free
Mesocolic



Internal features

Semicircular folds
No villi



VERMIFORM APPENDIX AND CECUM

Located in the right iliac fossa

Both parts are mostly

INTRAPERITONAL

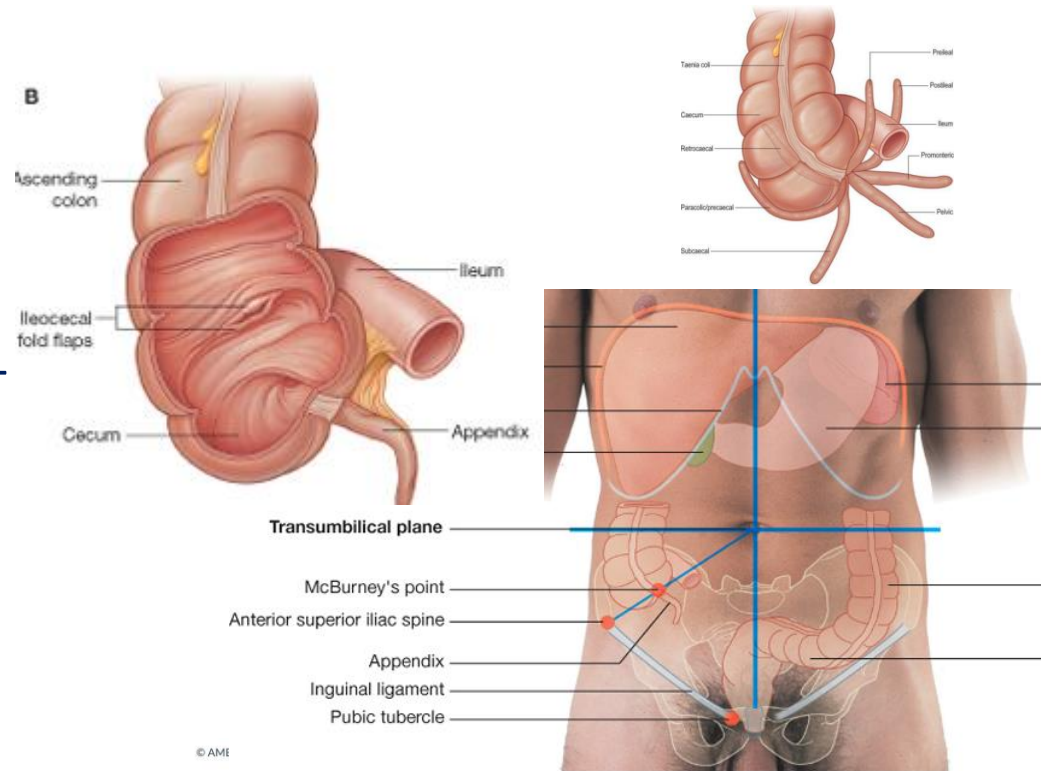
Ileocecal recesses (peritoneum)

Ileocecal orifice + Valve of Bauhin –
two lips (longitudinal muscle layer of
ileum)

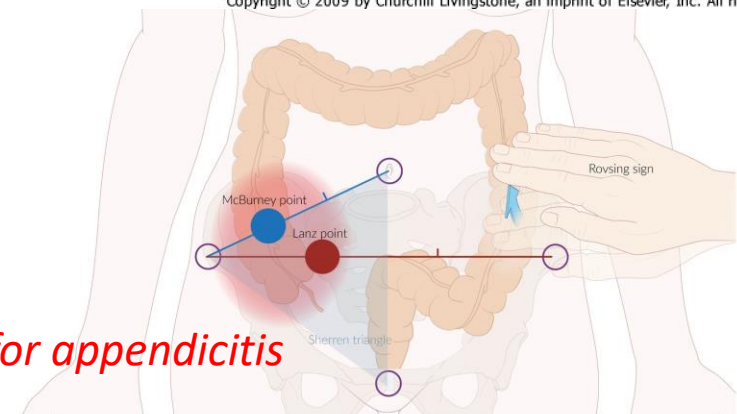
Vermiform appendix starts at the
merging of the 3 taeniae (McBurney
point) and ends at the Lanz's point
(approx 3-20 cm long)

RETROCECAL / RETROCOLIC
Great varieties in the position

Mesoappendix (if intraperitoneal)



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Pressure points for appendicitis

ASCENDING COLON

Continuation of the cecum

3 taeniae

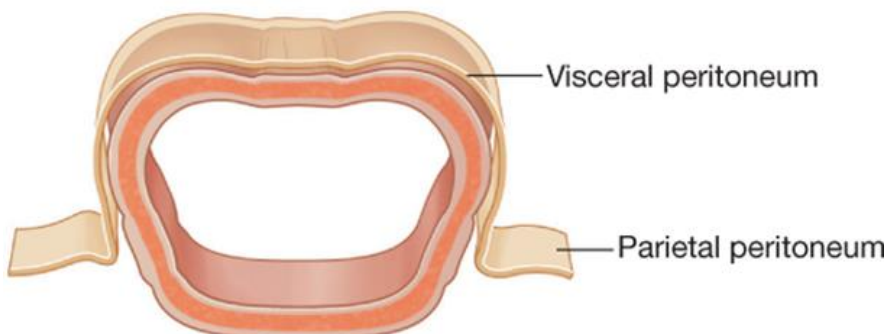
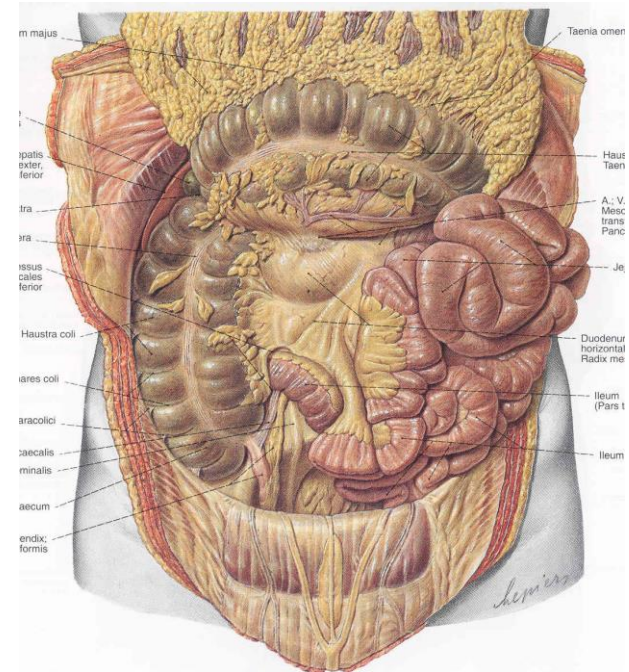
Partially intraperitoneal/retroperitoneal

Paracolic sulcus

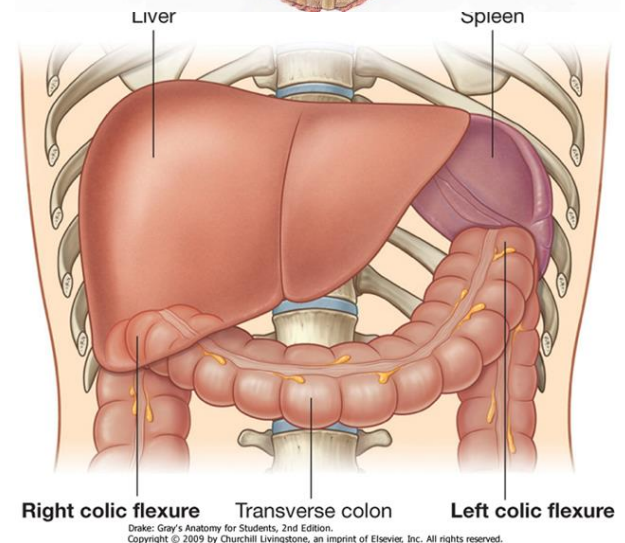
Position: lateral to the *midinguinal line*

- up to the *transpyloric plane*
- ends in the *right colic flexure*

Touches the posterior abdominal wall + right kidney



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Right colic flexure Transverse colon Left colic flexure

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TRANSVERSE COLON

Stretches from the **right colic flexure** to the **left colic flexure**
(right lumbar region, across the epigastrium to the left hypochondrium)
then continues downwards below the spleen

3 taeniae

Fully intraperitoneal

transverse mesocolon

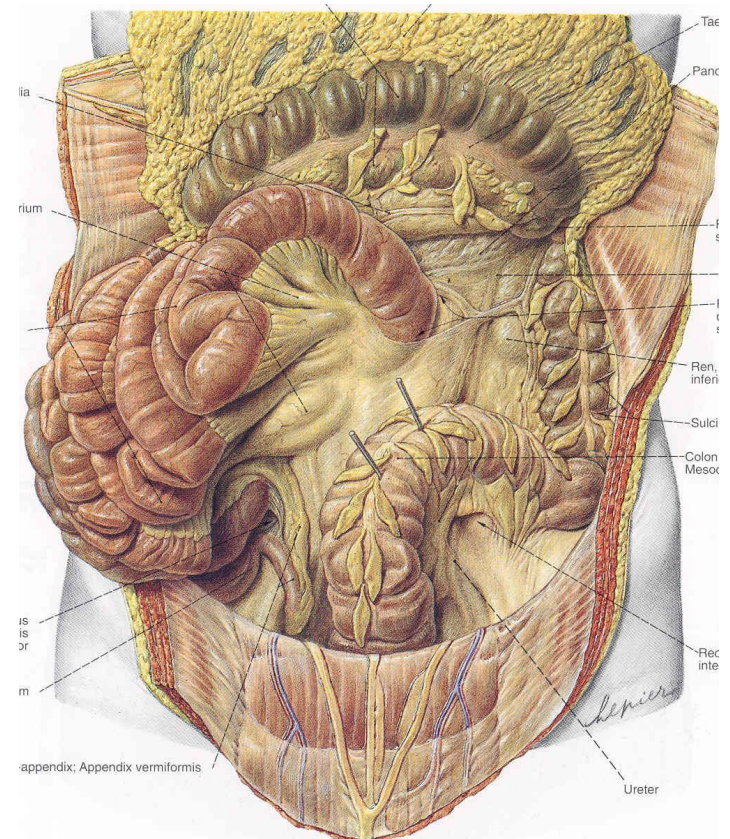
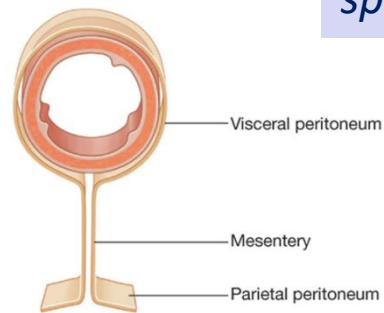
gastrocolic ligament (of the greater omentum)
connects to the greater curvature of the stomach

Syntopy of the right colic flexure

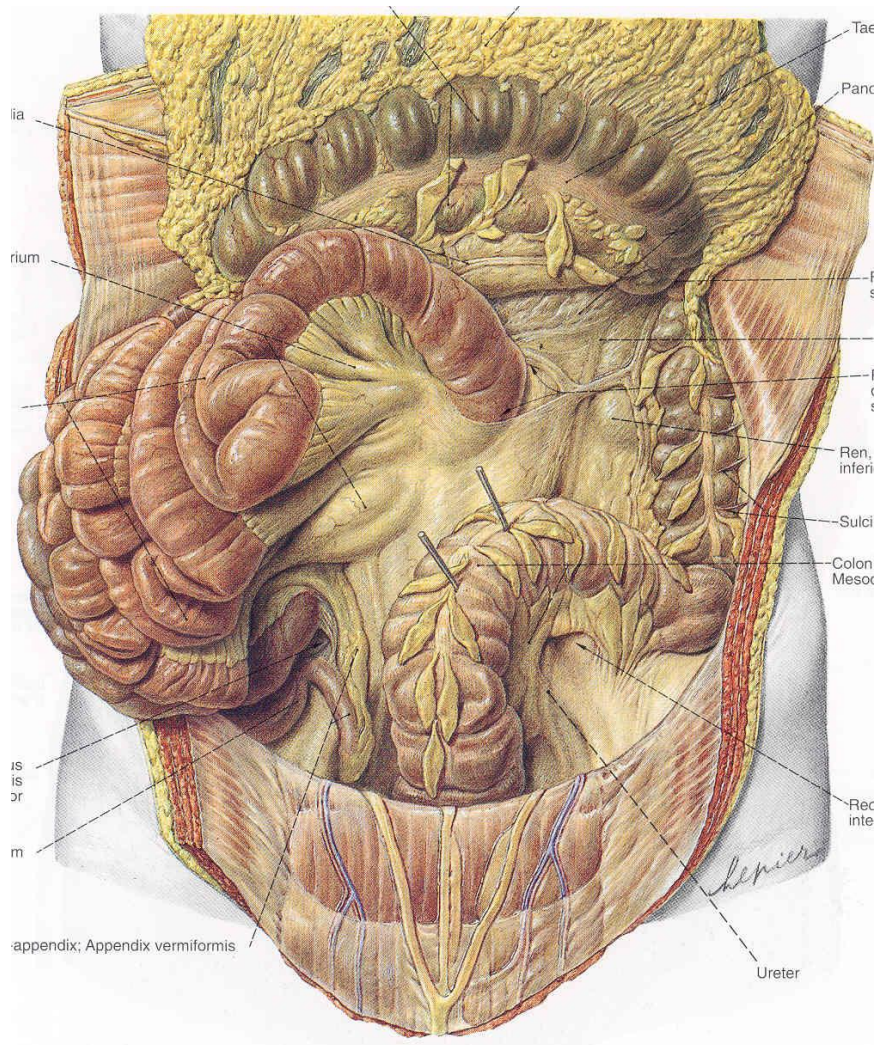
Right kidney
right lobe of liver (colic impression)
biliary vesicle
duodenum (desc part)

Syntopy of the left colic flexure

Lies higher than the right flexure
phrenicocolic ligament * 10-11th rib level
tail of pancreas
spleen (colic impression)



DESCENDING COLON



Continuation of the transverse colon
(heading down to the left iliac fossa)

Thinnest part of the large intestine
Develops from the hindgut

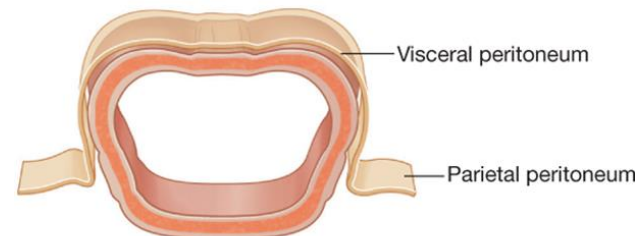
3 taeniae

Partially intraperitoneal/retroperitoneal

Paracolic sulcus

Position: lateral to the *midinguinal line*
- ends in the *sigmoid colon*

Touches the spleen, the posterior abdominal wall/diaphragm + left kidney



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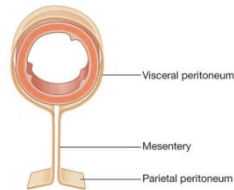
SIGMOID COLON

Descends from the left iliac fossa to the pelvis then ascends and again descends (**sigmoid looping**) to continue in the rectum at S3 level

Develops from the hindgut

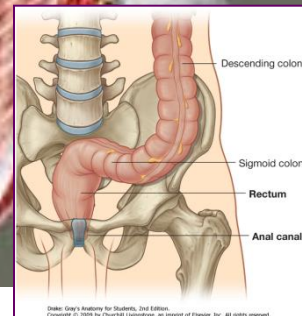
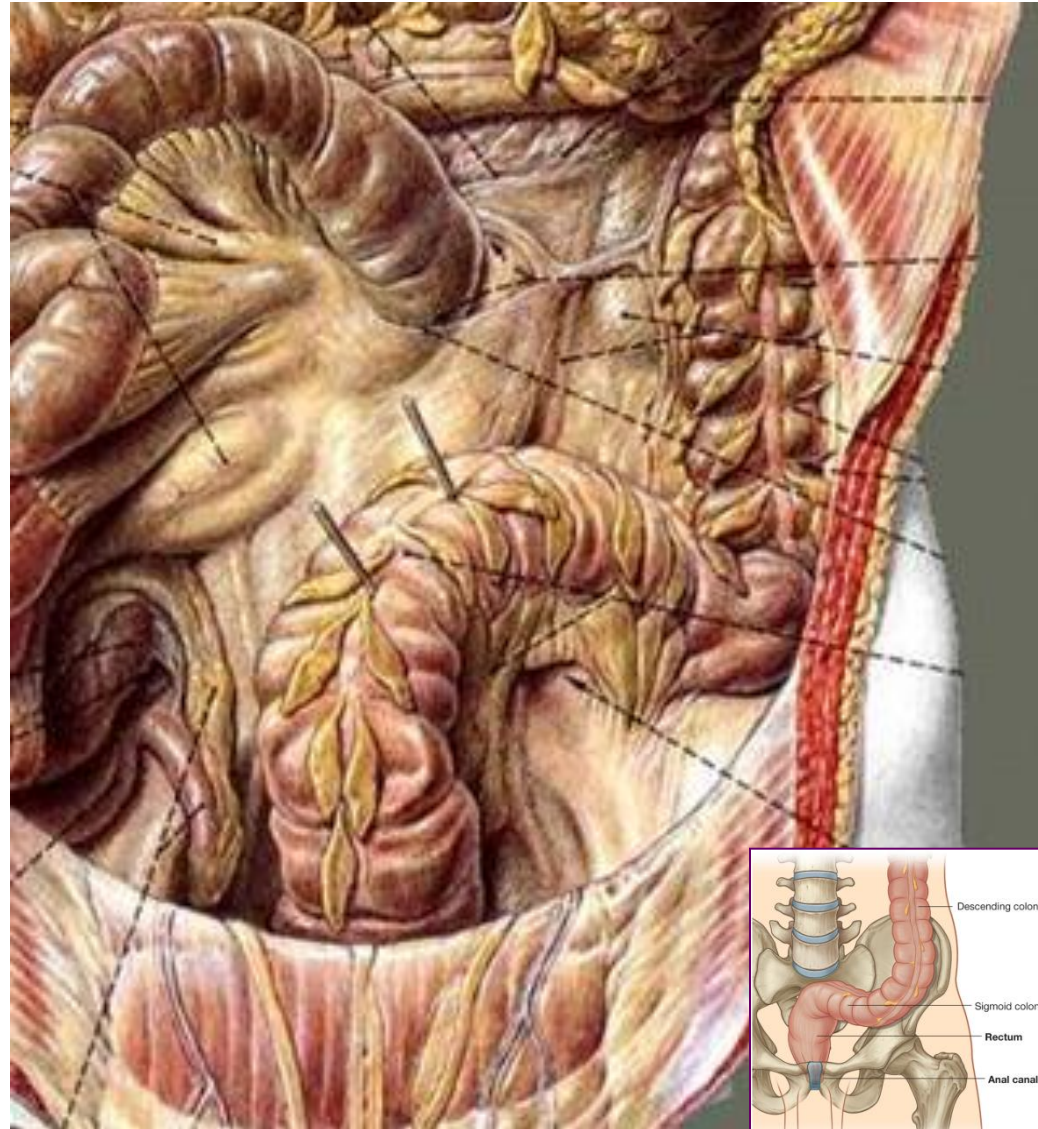
(3) → 2 taeniae

Intraperitoneal -
mesosigmoideum
mobile part of colon



Intersigmoid recess

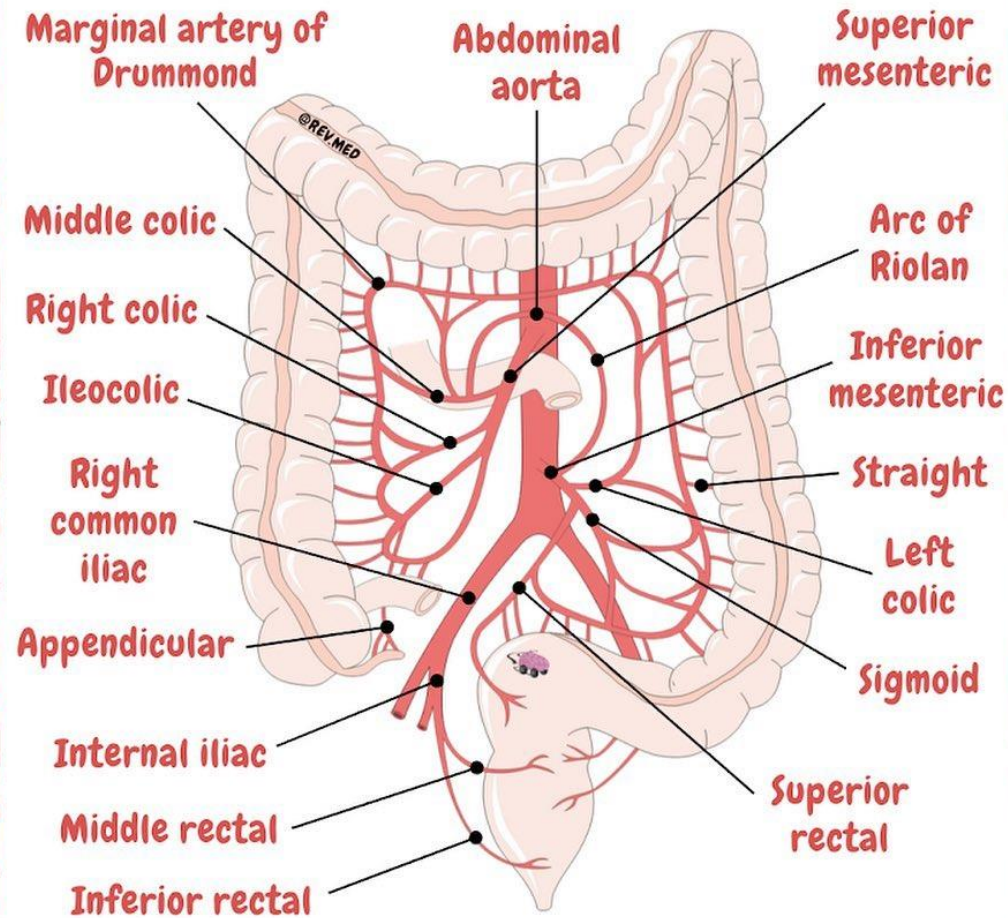
Touches the loops of the small intestine and the pelvic organs
(M: bladder /rectum
F: uterus/rectum)



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Large Intestine

BLOOD SUPPLY



SUPERIOR MESENTERIC ARTERY (L1)

Ileocolic artery

Supplies the cecum, vermiform appendix and the beginning of the ascending colon.

- *appendicular artery*
- *cecal arteries*

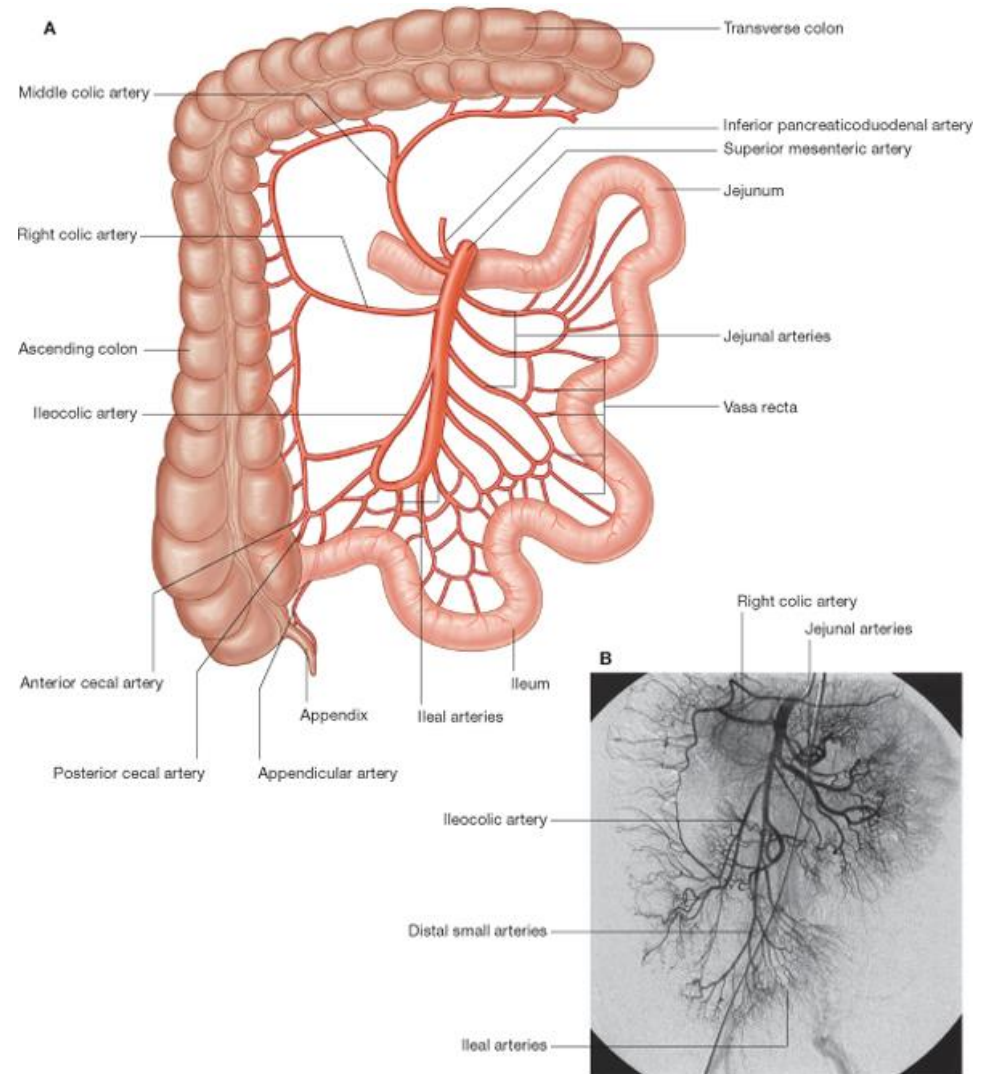
Right colic artery

Supplies the ascending colon and the right colic flexure

Middle colic artery

Supplies the right colic flexure and the transverse colon

(developmental border between midgut and hindgut)



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INFERIOR MESENTERIC ARTERY (L3)

Left colic artery

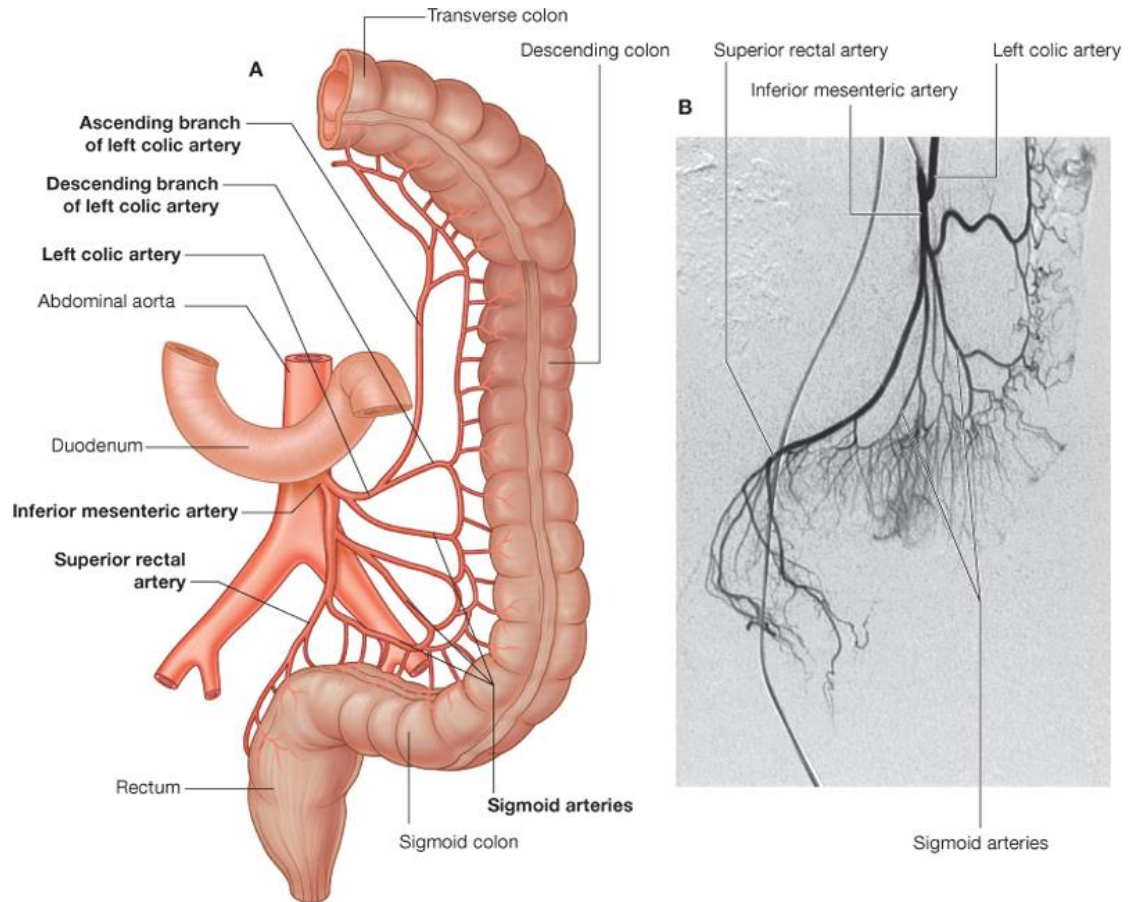
supplies part of the transverse colon, the left colic flexure and the descending colon

Sigmoid arteries

supplies the sigmoid colon (mesosigmoid)
2 -4 branches

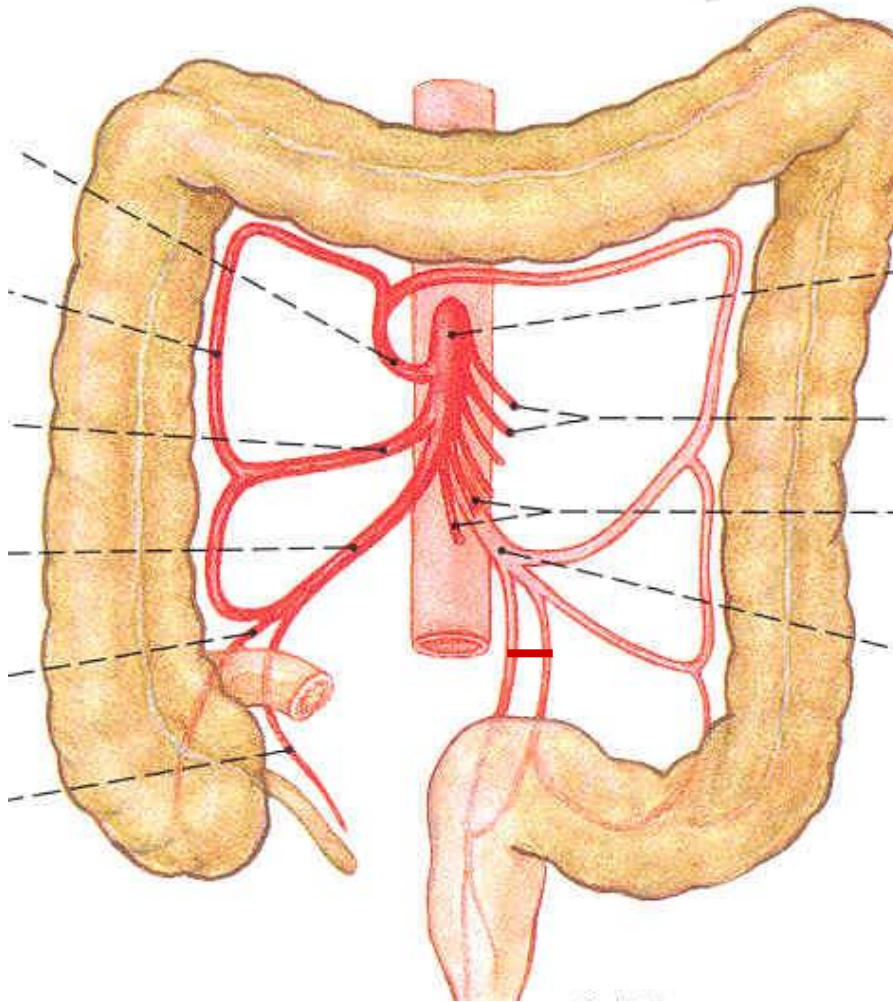
Superior rectal artery

Supplies the upper (larger part of the rectum (anastomoses with the middle rectal arteries – branch of the internal iliac)



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IMPORTANT ANASTOMOSES



ARCUS RIOLANI MAJOR (HALLER)

Drummond arch

secondary anastomosis

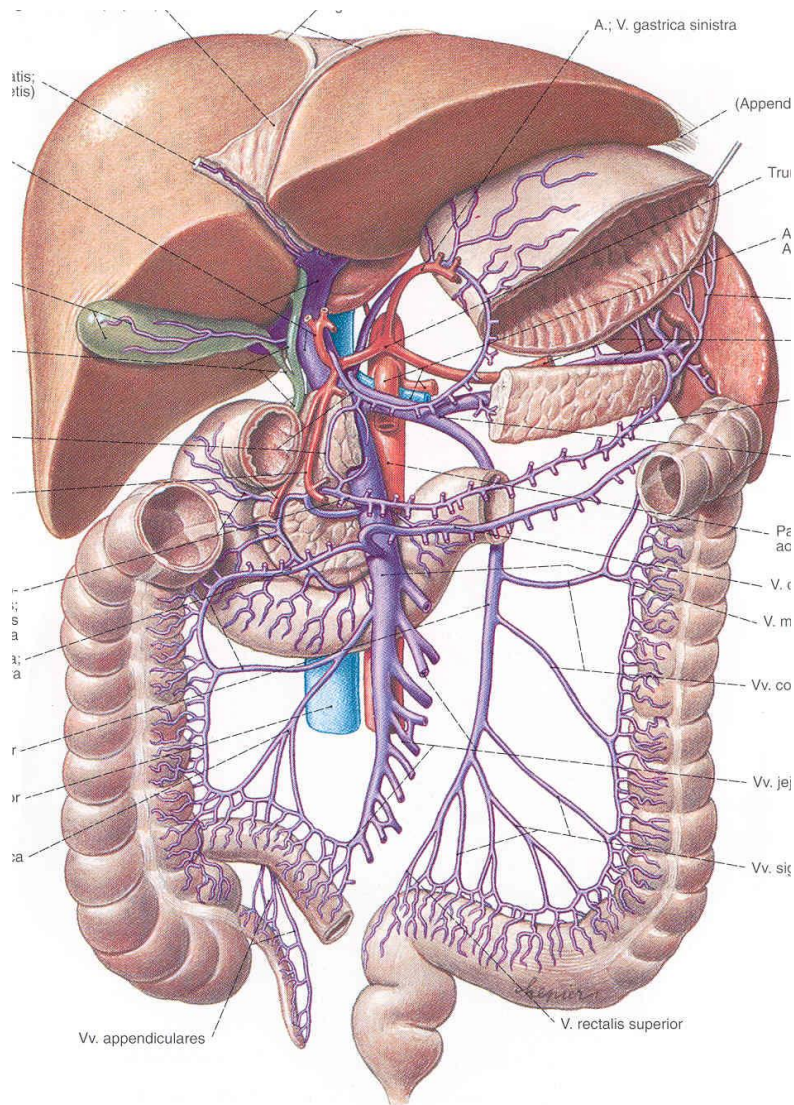
**Connects the middle and left
colic arteries (*embryological
border*)**

SUDECK'S POINT

***connects the superior rectal
artery and the last sigmoidal
branch***

***Anastomosis between the
superior rectal and middle
rectal arteries***

VENOUS DRAINAGE



Veins follow the course of arteries and aim for the portal vein

SUPERIOR MESENTERIC VEIN

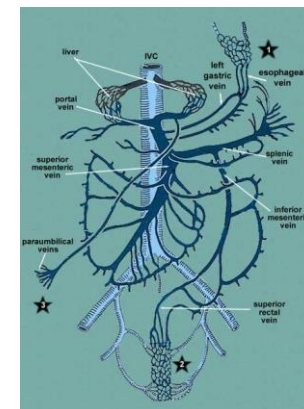
Drains the ileocolic, right and middle colic veins

Opens directly in the portal vein

INFERIOR MESENTERIC VEIN

Drains the left colic, sigmoid and superior rectal veins, opens either directly in the portal vein or indirectly, via the splenic vein

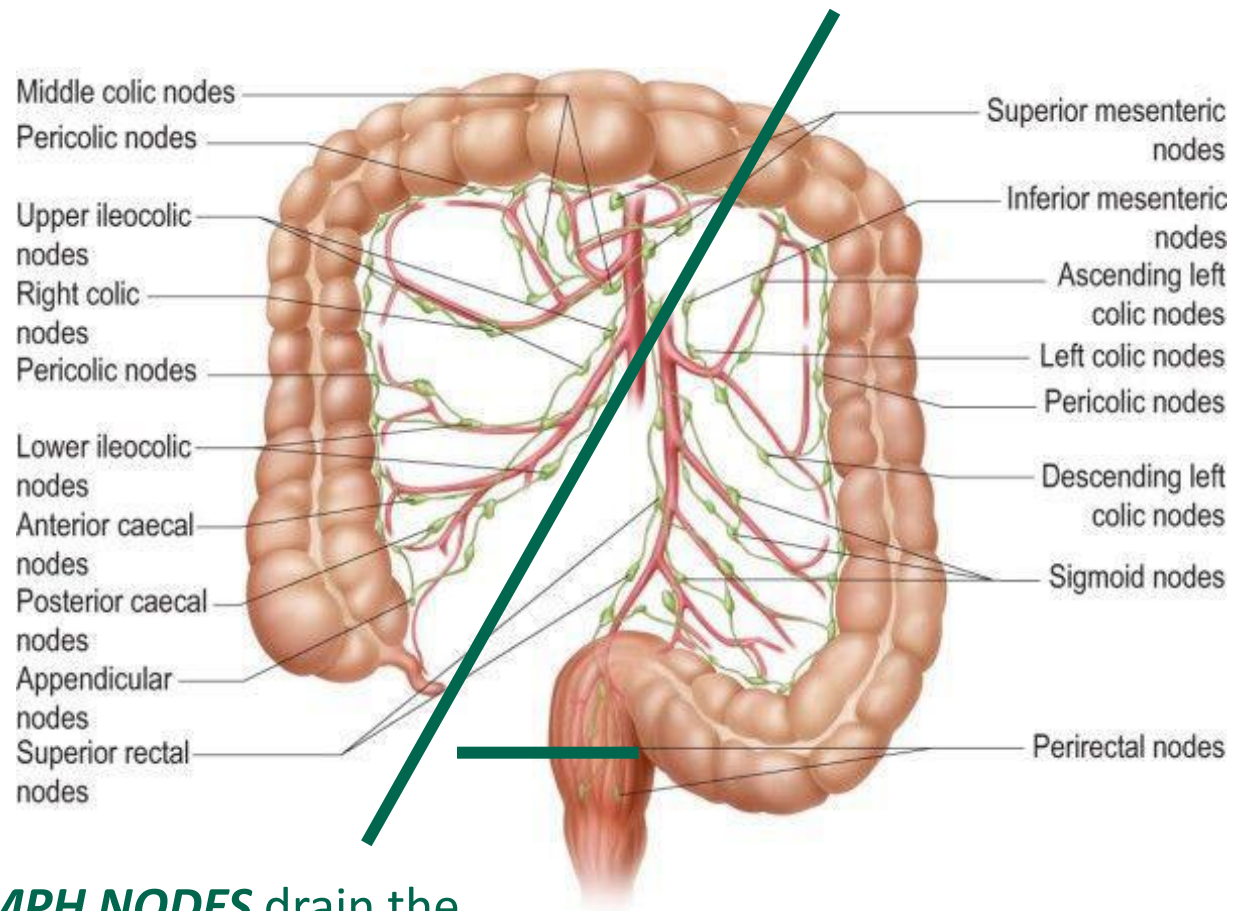
Please note the perirectal PORTOCAVAL ANASTOMOSIS



LYMPHATIC DRAINAGE

Lymph vessels run with the arteries

SUPERIOR MESENTERIC LYMPH NODES drain the cecum, ascending colon, transverse colon via the right and middle colic lymph nodes



INFERIOR MESENTERIC LYMPH NODES drain the *descending colon via the left colic lymph nodes*; the sigmoid colon via the sigmoidal lymph nodes; the upper 1/3 of the rectum

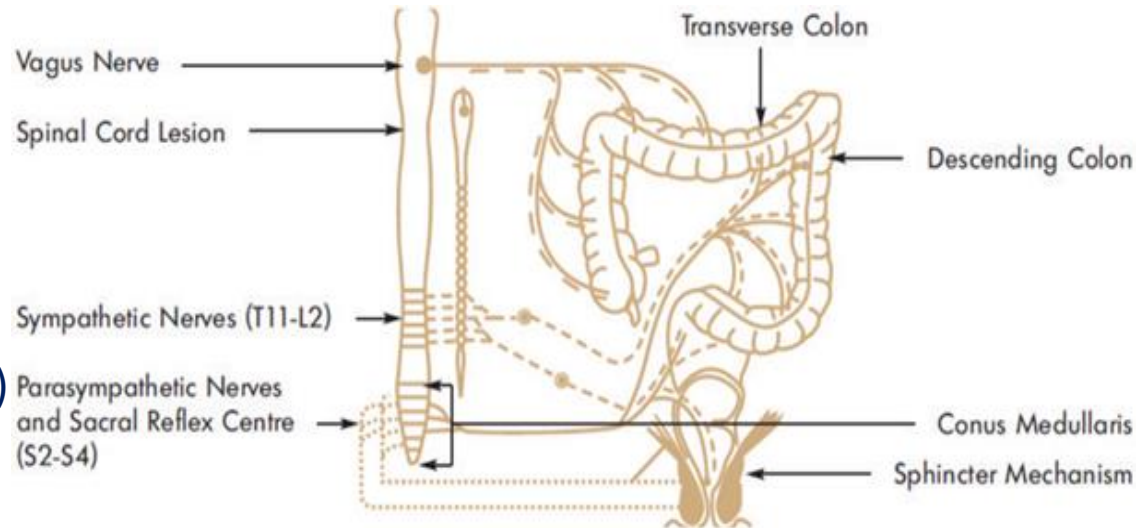
INTERNAL AND EXTERNAL ILIAC LYMPH NODES drain the inferior 2/3 of the rectum

INNERVATION

PARASYMPATHETIC INNERVATION
(secretomotor and motor to the colic muscles)

Vagus nerve innervates the colon to the splenic flexure (Cannon-Böhm's point).

Sacral parasympathetic fibres (S3-S5) innervate the descending colon, sigmoid colon and rectum.



SYMPATHETIC INNERVATION

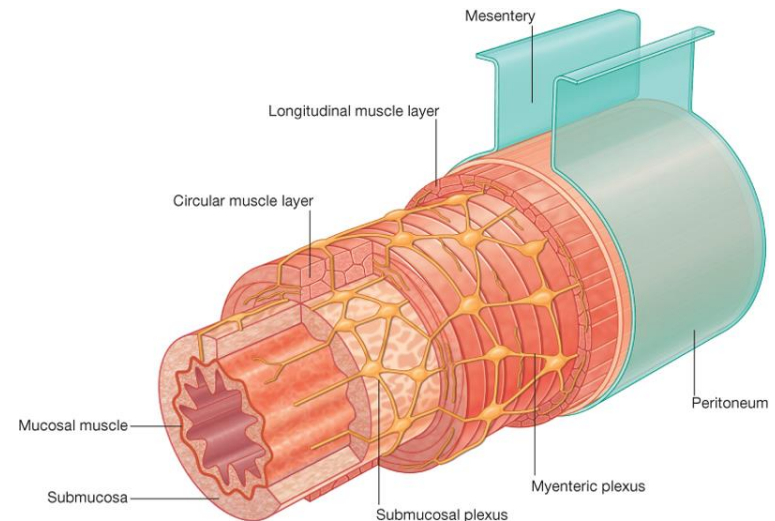
(vasoconstrictor and inhibitory to the colic muscles)

superior and inferior mesenteric plexus

greater and lesser splanchnic nerves (T6-11 ganglia)

Enteric nervous system:

Meissner's plexus and Auerbach's plexus



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